Quant edge 1.0 Final Challenge

A National STEM Quantitative Investment Research Competition

From Portfolio Theory to Value at Risk, Expected Shortfall and Extreme Value Theory An Applied Research

Challenge for STEM Undergraduates

Competition Briefing

Oct 2026

1. Executive Summary

The Quant*Edge* Challenge is a national competition that channels the analytical strength of Sri Lanka's leading STEM undergraduates - students trained in mathematics, statistics, engineering and computer science - into applied investment risk research. Teams of five build a working **Python** and **Streamlit** application that takes a portfolio of real market data from raw prices through to a defensible, risk-aware investment decision, using the same risk toolkit that professional asset managers, banks and regulators use: portfolio theory, Value-at-Risk (VaR), Expected Shortfall (ES), Extreme Value Theory (EVT) for tail risk.

The design deliberately starts from one simple, intuitive idea. Successful investing is not about maximizing returns, but about achieving sustainable returns while protecting capital and managing downside risk. From this foundation, increasingly rigorous statistical techniques are introduced so that every analytical tool builds on concepts teams have already learned to understand and justify.

2. Programme Objectives

The Challenge is designed to introduce students to the principles and practices that underpin modern investment and risk management, with content informed by the CFA Program curriculum and investmentindustry standards. Rather than focusing solely on maximizing returns, participants are encouraged to develop disciplined, evidence based investment strategies that balance return objectives with capital preservation and downside risk management. Through the application of quantitative techniques, coding, and financial analysis aligned with relevant CFA curriculum concepts and professional standards, students will gain practical experience in building transparent, reproducible solutions while developing skills that are directly relevant to careers in investment management, banking and financial risk.

3. Challenge Briefing and Investment Philosophy

Official Brief

Your team is the quantitative research division of an asset management firm competing for a mandate to manage a **USD 10 million** portfolio on behalf of a conservative institutional investor, The Serendib University Endowment Fund. It is a long-term investment pool established to support the university's academic mission, including scholarships, research initiatives, faculty development and strategic capital projects.

The client places a strong emphasis on **capital preservation, risk management and portfolio resilience during adverse market conditions**.

Investment Objective- Endowment seeks to preserve and grow its capital over the long term while maintaining a conservative risk profile. The Investment Committee expects any recommended portfolio to balance capital preservation, income generation, and sustainable real growth, with particular emphasis on resilience during periods of market stress.

The investment committee believes that;

"Avoiding large losses is more important than maximizing gains"

"A good investment decision is not the one with the highest expected return - it is the one whose downside you have actually measured, tested, and can defend"

Your team must build an end to end Python and Streamlit application that

1. constructs and recommends a portfolio from the prescribed investable universe using quantitative portfolio construction techniques;
2. measures and compares portfolio risk using Value-at-Risk (VaR), Expected Shortfall (ES), and Extreme Value Theory (EVT)-based tail risk estimates;
3. validates selected risk models through out of sample back testing
4. evaluates portfolio behavior under historical and hypothetical stress scenarios
5. delivers a **clear investment recommendation** supported by quantitative evidence and communicated through an investment committee memorandum.
6. **Teams have full discretion over their analytical approach, choice of risk metrics, lookback period, and portfolio construction method. The quality, transparency and justification of each team's decisions will form part of the assessment**

Teams are expected to determine what constitutes a suitable portfolio for the client's stated objectives and risk tolerance, justify their assumptions, and clearly communicate the limitations of their analysis. **The quality of the reasoning behind the recommendation is as important as the recommendation itself**.

Manager's Responsibility. Given the client' s profile, you must:

1. determine what the appropriate investment horizon, and asset allocation should be, and defend your recommendation. Teams may construct portfolios using a single fund or any combination of eligible funds.
2. Determine an appropriate return objective consistent with the client's mandate and risk tolerance.
3. Define and justify the **risk constraints** used in portfolio construction.
4. Demonstrate why the proposed portfolio is suitable for a conservative institutional investor.
5. Quantify the trade-offs between return-seeking and risk reduction.
6. Clearly articulate the assumptions underpinning your investment recommendation
7. Downside protection and Tail-risk management
8. Portfolio robustness under stress scenarios
9. Optional: Identification and assessment of other relevant macroeconomic, industry, or market-specific risks and their potential impact on the portfolio

Accordingly, the committee requires managers to demonstrate:

1. Diversification benefits
2. Robust risk measurement
3. Tail-risk awareness
4. Practical stress testing
5. Portfolio resilience during market crises

The Investment Committee will evaluate not only your expected returns but also your ability to explain:

1. What can go wrong
2. How bad losses could become
3. How confident you are in your risk estimates

5. Investable Universe

#	Ticker	Asset	Sector / Role
1	**SPY**	S&P 500 ETF	US Large-Cap Equities
2	**EFA**	iShares MSCI EAFE	International Developed Equities
3	**EEM**	iShares MSCI Emerging Markets	Emerging Market Equities
4	**AGG**	iShares US Aggregate Bond	US Investment-Grade Bonds
5	**GLD**	SPDR Gold Shares	Gold / Real Asset
6	**XLE**	Energy Select Sector SPDR	Energy / Commodities proxy
7	**XLK**	Technology Select Sector SPDR	Technology Sector

6. Technology Stack & Streamlit Deliverable

Python is the required language for all analysis. Streamlit is the required interface layer - the point is to translate a technical analysis into something a non-technical investment committee could actually use, not to produce a notebook. The Streamlit app must include, at minimum:

1. A sidebar to choose the confidence level (90/95/99%), the VaR methodology, and the backtest date range.
2. A portfolio panel showing weights, cumulative return, and the drawdown chart.
3. A risk dashboard showing VaR, ES, and the EVT-based tail estimate side by side at the same confidence level, so the gap between them is visually obvious.
4. A backtesting panel showing the Kupiec/Christoffersen results and a traffic-light indicator.

5. A stress-test panel showing portfolio impact under each crisis scenario.
6. A clearly labelled "Assumptions & Limitations" page - non-negotiable and explicitly graded - stating in plain language what the model does not capture.

Suggested (all free-tier) toolkit: pandas / NumPy / SciPy for computation, statsmodels or arch for stationarity/GARCH work, a dedicated EVT library (e.g., pyextremes) or a hand-rolled GPD maximumlikelihood fit, matplotlib/Plotly for charts, and Streamlit Community Cloud or a university-hosted server for deployment. Keeping the toolkit free-tier is itself a fairness requirement - no team should have an advantage because it can afford paid data or compute.

7. Required Deliverables

1. A public (or judge-accessible private) GitHub repository with clean, documented, reproducible code and a clear README explaining how to run the Streamlit app.
2. The Streamlit application itself, either deployed or runnable locally from the submitted repository **with one command**.
3. A 7–15-page technical report / investment memo covering methodology, data sources and cleaning steps, results, and - explicitly - assumptions and limitations.
4. A one-page executive "tear sheet" summarising the recommended portfolio, its risk metrics, and the recommendation, suitable for a real investment committee to read under five minutes.
5. For the final 5 selected teams, A 10–12 minute live pitch with a presentation deck to the judging panel followed by Q&A, in which the panel behaves as the client's investment committee.

8. Technical & Methodological Framework

Technical methodology is organised as five layers, each building on the one before it. Teams are required to implement every layer; how they implement it (which VaR method, which EVT threshold rule, which optimiser) is where genuine differentiation and judged "innovation" appears.

8.1 Layer 1 - Foundations: Portfolio Theory and Investment KPIs

Every team starts from Markowitz mean-variance portfolio construction and the Capital Asset Pricing Model.

Teams then **compute** and **interpret** the standard institutional performance KPIs such as; Sharpe Ratio, Sortino Ratio, Treynor Ratio, Jensen's Alpha, Information Ratio, Maximum Drawdown, Calmer Ratio and etc.

8.2 Layer 2 - Value-at-Risk

Teams implement and compare three VaR methodologies (Historical simulation, Parametric and Monte Carlo) on the same portfolio, at both 95% and 99% confidence, and are explicitly required to discuss where the three disagree and why:

Teams are expected to evaluate and justify the assumptions underlying the risk models they use, rather than applying standard techniques without critical assessment.

8.3 Layer 3 - Expected Shortfall (CVaR)

Teams must also report Expected Shortfall, giving suitable rationale for modelling for CVaR in addition to VaR.

8.4 Layer 4 - Extreme Value Theory (EVT)

Teams need to explore advanced methods for assessing extreme downside risk and to evaluate how different modelling approaches perform under stressed market conditions. Particular emphasis should be placed on understanding the limitations of conventional risk measures and demonstrating how alternative techniques can provide additional insight into rare but potentially severe loss events. Threshold selection in EVT must be justified, not chosen arbitrarily.

8.5 Layer 5 - Model Validation (Backtesting)

A risk model that has not been tested against reality is an opinion, not a measurement. Teams must backtest their chosen VaR model out-of-sample using:

- Kupiec's Proportion-of-Failures test
- Christoffersen's independence test
- The Basel traffic-light approach

8.6 From metrics to decisions - the practical investor layer

Every quantitative layer above must ultimately serve one output: a decision. Teams are graded on how well they translate the numbers into portfolio actions:

- Risk budgeting and position sizing - using each position's marginal VaR/ES contribution.
- Scenario and stress testing - applying historical crisis return paths (e.g., the 2008 Global Financial Crisis, the March 2020 COVID crash, and the 2022 rate-hiking drawdown) plus **one hypothetical scenario** the team designs itself, and reporting the resulting portfolio impact.
- The investment committee memo - a one-page, plain-English recommendation (accept the mandate as proposed / adjust the risk limit / reject) that a real institutional client's board could actually read and act on.

8.7 Use of AI and Coding Assistants

AI tools and coding assistants may be used to support technical implementation, debugging, research and software development. However, all material submitted must reflect the team’ s own human judgement, analysis and decisions.

Teams must disclose the following information when using AI:

Item	Required disclosure
AI tool(s) used	E.g., Claude, ChatGPT, GitHub Copilot
Purpose	E.g., debugging, Streamlit layout, code scaffolding, documentation support
Human validation	How the team tested, checked and revised generated output
Material AI-generated components	E.g., "Initial Streamlit dashboard structure was AI-assisted; risk calculations were independently reviewed and tested by the team."

- Explicitly disclose all AI tools used and the purpose for which they were used;
- Independently review, test and validate any AI-assisted output;
- **Understand and be able to explain the logic**, assumptions and limitations of all material submitted;
- Take full responsibility for the accuracy, reproducibility and interpretation of their code and results;
- Ensure that portfolio construction, risk interpretation, stress-test conclusions and final investment recommendations are developed and defended by the team.

AI tools must not be used as a substitute for the team’ s analytical judgement. Teams may not present an AI-generated recommendation, conclusion or investment strategy as their own without independently evaluating and substantiating it.

During judging, teams may be asked to explain any part of their code, methodology, analysis or recommendation. **Inability to explain or defend material aspects of the submission may result in a reduction in marks or disqualification.**

9. Data Sources & Data Governance

9.1 Approved data sources

- Yahoo Finance (via the yfinance Python package) - the primary source for global equity indices, ETFs, commodities, and FX, used for every team's core universe.
- FRED (Federal Reserve Economic Data) - for the risk-free rate proxy (e.g., 3-month U.S. T-bill yield) and macro series used in CAPM and stress-scenario construction.
- World Bank / IMF open data - for macro context in the investment memo, where relevant.

All teams work from the same permitted source list; using an undisclosed or proprietary data source is a rubric violation, not a bonus - reproducibility by the judges is a graded requirement.

9.2 Mandatory data-cleaning discipline

- Use adjusted close prices so dividends and stock splits do not distort return series.
- Align trading calendars across exchanges before computing portfolio-level returns; document how holidays and non-overlapping trading days were handled.
- Convert to log returns for statistical work and clearly state where simple returns are used instead and why.
- Flag, don't silently delete, suspected data errors (stale/flat-lined prices, structural breaks, implausible jumps, large negatives) - outlier handling must be documented, not hidden.
- Run a basic stationarity check (e.g., an Augmented Dickey-Fuller test) on the return series used for modelling, and note the result.
- Every cleaning step must be reproducible from the submitted code - no manual, undocumented edits to a downloaded file.

10. Judging Criteria & Rubric

The rubric is deliberately weighted so that technical correctness and decision-usefulness together outweigh a beautifully presented but statistically wrong analysis should not win.

Criterion	Weight	What is being assessed
Investment decisionmaking	30%	Does the analysis lead to a sound, actionable, client-appropriate recommendation, not just a set of numbers. Analysis depth and assessment of all relevant variables (including optional items) supporting the investment decision.
Technical rigor & correctness	25%	Correct assessment, implementation and interpretation of KPIs, VaR, ES, EVT, any optional macro/ industry variables, and statistically sound backtesting
Software & interface quality	15%	Streamlit usability, front-end design, code structure, documentation, and reproducibility from the submitted repository
Innovation & ambition	15%	Well-justified extensions: e.g., regime-conditioned risk, optional risk overlays, alternative EVT threshold rules
Communication & defence	15%	Clarity of the written memo and live pitch; quality of answers under judges' questioning